

STORMO.IO

PWA Implementation Report

Progressive Web App — Cross-Platform Compliance & Device Coverage

Field	Detail
Project	Stormo.io — AI Marketing Manager
Report Version	1.0
Date	June 2026
Prepared By	Muhammad Ismaeel
Platforms	iOS · Android · Desktop
Reference Guide	Stormo PWA iOS Technical Guide (June 2026)
Status	All Requirements Met — PWA Live

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Section 1 → iOS PWA Compliance

Reference: Stormo PWA iOS Technical Guide — "All six items must be implemented from the start of the Next.js build."

All six required items from the iOS Technical Guide have been implemented. Status for each is verified below.

1.1 — Apple Meta Tags in `layout.tsx`

Requirement: iOS requires Apple-specific meta tags in addition to the standard web app manifest. Without these, Stormo renders as a plain browser bookmark on the merchant's iPhone home screen — no splash screen, wrong icon, browser chrome visible.

Requirement	Status	Implementation
<code>appleWebApp.capable = true</code>	 Done	<code>app/layout.tsx</code> — metadata export
<code>appleWebApp.title = 'Stormo'</code>	 Done	<code>app/layout.tsx</code> — metadata export
<code>appleWebApp.statusBarStyle</code>	 Done	Set to <code>black-translucent</code> (content extends behind status bar)
<code>apple-touch-icon</code> 180×180	 Done	<code>/icons/apple-touch-icon.png</code> — realfavicongenerator.net
Startup images linked	 Done	22 portrait splash screens — all devices covered
Icons have no pre-rounded corners	 Done	iOS applies corner rounding automatically

Checklist:

-  `appleWebApp.capable: true` present in metadata
-  `appleWebApp.statusBarStyle: 'black-translucent'` — content shows behind Dynamic Island
-  `appleWebApp.title: 'Stormo'` — displays under home screen icon
-  `apple-touch-icon.png` (180×180) present in `/public/icons/`
-  `startupImage` array populated with correct device media queries
-  No pre-rounded corners on apple-touch-icon (iOS rounds automatically)

1.2 — Web App Manifest (`manifest.json`)

Requirement: Incorrect or missing fields cause Stormo to open in the browser instead of as a standalone app after installation. `display` must be `standalone` — the only mode iOS supports that removes Safari browser chrome.

Field	Required Value	Current Value	Status
<code>name</code>	Full app name	<code>Stormo.io – AI Marketing Manager</code>	 Done
<code>short_name</code>	Home screen label	<code>Stormo</code>	 Done
<code>start_url</code>	<code>/</code> or <code>/dashboard</code>	<code>/</code>	 Done
<code>scope</code>	<code>/</code>	<code>/</code>	 Done
<code>display</code>	<code>standalone</code> only	<code>standalone</code>	 Done
<code>background_color</code>	Brand dark	<code>#1A1A1A</code>	 Done
<code>theme_color</code>	Brand primary	<code>#E8621A</code>	 Done
<code>icons</code> — 192×192 any	Required	<code>web-app-manifest-192x192.png</code>	 Done

Field	Required Value	Current Value	Status
<code>icons</code> — 512×512 any	Required	<code>web-app-manifest-512x512.png</code>	✓ Done
<code>icons</code> — maskable	Required for Android	192 + 512 maskable declared	✓ Done
<code>orientation</code>	any	any	✓ Done
<code>shortcuts</code>	Optional, added	3 shortcuts: Action, Content, Outreach	✓ Done

Checklist:

- ✓ `display: "standalone"` — not `fullscreen` or `browser`
- ✓ Both 192px and 512px `any` icons declared
- ✓ Maskable icons declared (192px + 512px)
- ✓ `background_color` matches splash screen background (`#1A1A1A`)
- ✓ `theme_color` matches brand orange (`#E8621A`)
- ✓ `scope: "/"` set — app controls entire origin
- ✓ `shortcuts` added — 3 deep-link shortcuts for Android home screen long-press

1.3 — Service Worker — iOS Caching Strategy

Requirement: iOS deletes a PWA's local cache after 7 days of inactivity. Service worker must cache the app shell on every launch and always fetch fresh API data. Merchants who return after 8+ days must get a fast, reliable load — a blank screen damages trust before the product delivers value.

Caching Strategy	URL Pattern	Behaviour	Status
Cache-First	<code>/_next/static/</code>	JS/CSS chunks served from cache instantly	✓ Done
Cache-First	Static images & fonts	Assets cached 30 days	✓ Done
Network-First	Page navigation	Fresh HTML, offline fallback to cache	✓ Done
Network-Only	<code>/api/</code> routes	API data always live — never stale	✓ Done
Precache on install	Offline page + manifest	Available immediately on first load	✓ Done
Stale-While-Revalidate	Images + fonts	Instant display, background refresh	✓ Done

Checklist:

- ✓ Custom `public/sw.js` registered via `PwaProvider.tsx` on mount
- ✓ `skipWaiting()` called on install — new SW activates immediately
- ✓ `clients.claim()` called on activate — existing pages controlled immediately
- ✓ Old caches purged on activate — no stale cache accumulation
- ✓ `public/offline.html` exists and is precached — works without network
- ✓ `sw.js` served with `Cache-Control: max-age=0, must-revalidate` — always fresh
- ✓ `Service-Worker-Allowed: /` header set — SW controls full origin scope
- ✓ API routes (`/api/`) are network-only — no stale merchant data ever served

1.4 — iOS Install Prompt — Custom UI Component

Requirement: iOS does not allow browsers to trigger a native install prompt. There is no install button built into Safari. The merchant must manually tap Share → Add to Home Screen. Without a custom prompt guiding them through this, most merchants will never install Stormo as a PWA.

Requirement	Status	Implementation
Custom install prompt component	✓ Done	<code>components/pwa/PwaProvider.tsx</code>
iOS device detection	✓ Done	<code>`/iphone`</code>
Standalone mode detection	✓ Done	<code>window.navigator.standalone</code> + <code>display-mode</code>
No prompt shown if already installed	✓ Done	Exits early if standalone mode detected
Step-by-step guide shown	✓ Done	3-step: Share icon → Add to Home Screen → Add
Auto-shown after 4 seconds	✓ Done	<code>setTimeout</code> 4000ms on page load
Dismissable, remembered per session	✓ Done	<code>sessionStorage</code> dismissal key
Android install button (separate flow)	✓ Done	<code>beforeinstallprompt</code> event → native prompt

Checklist:

- ✓ iOS-specific UI shown (no native prompt exists on iOS)
- ✓ Android/Chrome `beforeinstallprompt` captured and shown as install button
- ✓ Already-installed users see no prompt (standalone detection)
- ✓ Prompt dismisses cleanly and does not reappear same session
- ✓ Step 1: Tap Share icon in Safari
- ✓ Step 2: Scroll down and tap "Add to Home Screen"
- ✓ Step 3: Tap "Add" in the top-right corner
- ✓ Prompt works on iPhone and iPad — Safari only note displayed

1.5 — Viewport Fit (Notch / Dynamic Island Support)

Requirement: `viewport-fit=cover` must be set so the app content extends behind the iOS status bar. Required for iPhone X and later (notch, Dynamic Island). Without this, there is a white band at the top of the screen in standalone mode.

Requirement	Status	Implementation
<code>viewport-fit=cover</code>	✓ Done	<code>app/layout.tsx</code> — <code>viewport</code> export
<code>viewportFit: 'cover'</code>	✓ Done	Next.js <code>Viewport</code> type, <code>viewportFit</code> field
Safe area CSS variables	✓ Done	<code>app/globals.css</code> — <code>--safe-top/bottom/left/right</code>
<code>env(safe-area-inset-*)</code> used	✓ Done	PWA banner, offline page use safe area padding
<code>.pwa-safe-top</code> utility class	✓ Done	Applied when <code>display-mode: standalone</code>
Offline page safe area padding	✓ Done	<code>public/offline.html</code> uses <code>env(safe-area-inset-*)</code>

Checklist:

- ✓ `viewportFit: 'cover'` in `viewport` export
- ✓ `--safe-top: env(safe-area-inset-top, 0px)` CSS variable

- ☒ `--safe-bottom: env(safe-area-inset-bottom, 0px)` CSS variable
- ☒ `--safe-left: env(safe-area-inset-left, 0px)` CSS variable
- ☒ `--safe-right: env(safe-area-inset-right, 0px)` CSS variable
- ☒ `.pwa-safe-top` and `.pwa-safe-bottom` utility classes available
- ☒ `-webkit-tap-highlight-color: transparent` — no blue flash on tap

1.6 — iOS Splash Screens (All Devices)

Requirement: Apple requires a unique splash screen for every device resolution. Without the correct `media` query match, iOS shows a white screen during app load instead of the branded splash. This is especially visible on first launch.

Device Group	File	Dimensions	Status
iPhone 17 Pro Max / 16 Pro Max	<code>iPhone_17_Pro_Max__iPhone_16_Pro_Max_portrait.png</code>	1320×2868	☒ Done
iPhone 17 Pro / 17 / 16 Pro	<code>iPhone_17_Pro__iPhone_17__iPhone_16_Pro_portrait.png</code>	1206×2622	☒ Done
iPhone Air	<code>iPhone_Air_portrait.png</code>	1260×2736	☒ Done
iPhone 16 Plus / 15 Pro Max / 14 Pro Max	<code>iPhone_16_Plus__iPhone_15_Pro_Max..._portrait.png</code>	1290×2796	☒ Done
iPhone 16 / 15 Pro / 15 / 14 Pro	<code>iPhone_16__iPhone_15_Pro__iPhone_15__iPhone_14_Pro_portrait.png</code>	1179×2556	☒ Done
iPhone 14 Plus / 13 Pro Max / 12 Pro Max	<code>iPhone_14_Plus__iPhone_13_Pro_Max..._portrait.png</code>	1284×2778	☒ Done

Device Group	File	Dimensions	Status
iPhone 17e / 16e / 14 / 13 / 12	iPhone_17e__iPhone_16e__iPhone_14_..._portrait.png	1170×2532	 Done
iPhone 13 mini / 12 mini / 11 Pro / X	iPhone_13_mini__iPhone_12_mini_..._portrait.png	1125×2436	 Done
iPhone 11 Pro Max / XS Max	iPhone_11_Pro_Max__iPhone_XS_Max_portrait.png	1242×2688	 Done
iPhone 11 / XR	iPhone_11__iPhone_XR_portrait.png	828×1792	 Done
iPhone 8 Plus / 7 Plus / 6s Plus	iPhone_8_Plus__iPhone_7_Plus_..._portrait.png	1242×2208	 Done
iPhone 8 / 7 / 6s / SE 2nd gen	iPhone_8__iPhone_7__iPhone_6s__4.7__iPhone_SE_portrait.png	750×1334	 Done
iPhone SE 1st gen / iPod	4__iPhone_SE__iPod_touch_..._portrait.png	640×1136	 Done
iPad Pro M4 13"	13__iPad_Pro_M4_portrait.png	2064×2752	 Done
iPad Pro 12.9"	12.9__iPad_Pro_portrait.png	2048×2732	 Done
iPad Pro M4 11"	11__iPad_Pro_M4_portrait.png	1668×2420	 Done
iPad Pro 11"	11__iPad_Pro__10.5__iPad_Pro_portrait.png	1668×2388	 Done

Device Group	File	Dimensions	Status
iPad Air 10.9"	10.9__iPad_Air_portrait.png	1640×2360	✓ Done
iPad Air 10.5"	10.5__iPad_Air_portrait.png	1668×2224	✓ Done
iPad 10.2"	10.2__iPad_portrait.png	1620×2160	✓ Done
iPad Mini 8.3"	8.3__iPad_Mini_portrait.png	1488×2266	✓ Done
iPad 9.7" (legacy)	9.7__iPad_Pro__7.9__iPad_mini_..._portrait.png	1536×2048	✓ Done

Total: 22 portrait splash screens — complete device coverage from iPhone SE (2013) to iPhone 17 Pro Max (2025)

Checklist:

- ✓ All 13 iPhone models covered (portrait)
- ✓ All 9 iPad models covered (portrait)
- ✓ All splash screens placed in public/splash/
- ✓ All media queries in layout.tsx use correct device-width, device-height, -webkit-device-pixel-ratio
- ✓ orientation: portrait added to every media query
- ✓ Every file path in layout.tsx verified to exist on disk
- ✓ Splash background colour matches background_color in manifest (#1A1A1A)

Section 2 → Android & Chrome PWA Checklist

Requirement	Status	Note
Web App Manifest present and valid	✓ Done	/manifest.json served with correct Content-Type
display: standalone in manifest	✓ Done	Chrome uses manifest; iOS uses apple meta tags
192×192 icon in manifest	✓ Done	web-app-manifest-192x192.png
512×512 icon in manifest	✓ Done	web-app-manifest-512x512.png
Maskable icon declared	✓ Done	192 + 512 with purpose: "maskable"
Service worker registered	✓ Done	PwaProvider.tsx registers on mount
beforeinstallprompt captured	✓ Done	Deferred, shown as "Add to Home Screen" button
Install prompt — dismissable	✓ Done	Dismiss stored in sessionStorage
Offline page	✓ Done	public/offline.html — Stormo branded

Requirement	Status	Note
<code>theme_color</code> in manifest	✓ Done	<code>#E8621A</code> — matches browser tab colour on Android
<code>background_color</code> in manifest	✓ Done	<code>#1A1A1A</code> — splash background while app loads
Shortcuts (long-press home screen icon)	✓ Done	3 shortcuts: Dashboard, Content, Outreach
HTTPS required for SW registration	✓ Ready	Vercel deployment uses HTTPS automatically

Checklist:

- ✓ Manifest linked in `<head>` via `metadata.manifest`
- ✓ Service worker scope is `/` — controls full origin
- ✓ Install button shown via `beforeinstallprompt` (Android/Chrome)
- ✓ User accepting install prompt confirmed in console log
- ✓ Offline fallback shows branded Stormo page, not browser default
- ✓ `shortcuts` allow fast access to Dashboard, Content, Outreach from home screen

Section 3 → Desktop PWA Checklist

Requirement	Status	Note
Installable via browser address bar	✓ Ready	Chrome/Edge show install icon in address bar if manifest valid
512×512 icon for high-DPI displays	✓ Done	<code>web-app-manifest-512x512.png</code>
SVG icon for any resolution	✓ Done	<code>favicon.svg</code> in manifest
<code>window</code> mode opens app standalone	✓ Done	<code>display: standalone</code> in manifest
Service worker runs on desktop	✓ Done	No platform restriction — registered across all browsers
Offline page works on desktop	✓ Done	Same <code>offline.html</code> served
Install banner (desktop variant)	✓ Done	PwaProvider shows 320px right-anchored banner on desktop
<code>overscroll-behavior: none</code>	✓ Done	<code>globals.css</code> — prevents bounce-scroll on macOS

Checklist:

- ✓ Valid manifest — Chrome/Edge show install prompt in address bar
- ✓ SVG icon scales perfectly at any desktop resolution
- ✓ Desktop install banner positioned bottom-right (320px wide)
- ✓ Offline mode functional on desktop browsers
- ✓ `overscroll-behavior: none` prevents body scroll bounce on macOS trackpad

Section 4 → Core Technical Requirements

4.1 — Files Delivered

File	Location	Purpose
<code>manifest.json</code>	<code>public/manifest.json</code>	PWA identity — name, icons, display, shortcuts
<code>sw.js</code>	<code>public/sw.js</code>	Service worker — caching, offline, push events
<code>offline.html</code>	<code>public/offline.html</code>	Branded offline fallback page
<code>PwaProvider.tsx</code>	<code>components/pwa/</code>	SW registration + install prompt (iOS + Android)
<code>icon.svg</code>	<code>public/icons/</code>	SVG icon for any-size usage
PWA meta in <code>layout.tsx</code>	<code>app/layout.tsx</code>	All <code><head></code> meta tags for iOS + Android
SW headers in <code>next.config.ts</code>	<code>next.config.ts</code>	Cache-Control + Service-Worker-Allowed headers
Safe area CSS	<code>app/globals.css</code>	<code>env(safe-area-inset-*)</code> variables + utilities

4.2 — HTTP Headers

Route	Header	Value
<code>/sw.js</code>	Cache-Control	<code>public, max-age=0, must-revalidate</code>
<code>/sw.js</code>	Service-Worker-Allowed	<code>/</code>
<code>/sw.js</code>	X-Content-Type-Options	<code>nosniff</code>
<code>/manifest.json</code>	Cache-Control	<code>public, max-age=0, must-revalidate</code>
<code>/manifest.json</code>	Content-Type	<code>application/manifest+json</code>
<code>/offline.html</code>	Cache-Control	<code>public, max-age=0, must-revalidate</code>

4.3 — Viewport Configuration

Setting	Value	Reason
<code>themeColor</code>	<code>#E8621A</code>	Browser tab / status bar tint
<code>width</code>	<code>device-width</code>	Standard responsive behaviour
<code>initialScale</code>	<code>1</code>	Default zoom
<code>maximumScale</code>	<code>5</code>	Allow pinch-to-zoom (accessibility)
<code>viewportFit</code>	<code>cover</code>	Content behind iOS notch / Dynamic Island

Section 5 → Icon & Asset Inventory

5.1 — Icons in `public/icons/`

File	Size	Usage
<code>apple-touch-icon.png</code>	180×180	iOS home screen icon (primary)
<code>favicon-96x96.png</code>	96×96	Browser favicon, PWA shortcuts icon

File	Size	Usage
web-app-manifest-192x192.png	192×192	Android home screen + manifest any / maskable
web-app-manifest-512x512.png	512×512	Android splash + manifest any / maskable
favicon.svg	Scalable	Browser tab, desktop PWA any -resolution
favicon.ico	Multi-size	Legacy browser fallback

5.2 — Favicon & Root Icons

File	Location	Usage
favicon.png	public/	Primary browser favicon
stormo-logo.png	public/	Source asset used in app UI

5.3 — Manifest Icon Declaration Summary

Purpose	Files Declared	Sizes
any	favicon-96x96, apple-touch-icon, web-app-manifest × 2	96, 180, 192, 512
maskable	web-app-manifest-192x192, web-app-manifest-512x512	192, 512

Checklist:

- ☒ At least one 192×192 any icon in manifest
- ☒ At least one 512×512 any icon in manifest
- ☒ At least one maskable icon — required for Android Adaptive Icons
- ☒ apple-touch-icon.png present for iOS home screen
- ☒ favicon.ico present for legacy browsers
- ☒ SVG icon present for scalable display
- ☒ All icon paths in manifest verified to exist on disk

Section 6 → Splash Screen Device Coverage

Total files in public/splash/: 44 files (22 portrait + 22 landscape) Used in layout.tsx: 22 portrait files — all with correct orientation: portrait media query

Platform	Models Covered	Files	Status
iPhone (modern)	17 Pro Max, 17 Pro, 17, Air, 16 Pro Max, 16 Pro	6	☒ Done
iPhone (current)	16 Plus, 16, 15 Pro Max, 15 Plus, 15, 14 Pro Max, 14 Pro, 14	5	☒ Done
iPhone (older)	13 series, 12 series, 11 series, XS, X, 8, 7, 6, SE	7	☒ Done
iPad Pro	M4 13", M4 11", 12.9", 11"	4	☒ Done
iPad Air	10.9", 10.5"	2	☒ Done
iPad / Mini	10.2", 9.7", Mini 8.3"	3	☒ Done
Total	All iOS/iPadOS devices (2013–2025)	22	☒ Done

Section 7 → Service Worker Feature Matrix

Feature	Implemented	Strategy	Cache Name
Next.js static JS/CSS chunks	✓ Yes	Cache-First	<code>stormo-static-v1</code>
Static images & fonts	✓ Yes	Stale-While-Revalidate	<code>stormo-static-v1</code>
Page navigation (HTML)	✓ Yes	Network-First + offline fallback	<code>stormo-pages-v1</code>
API routes (<code>/api/*</code>)	✓ Yes	Network-Only	N/A — never cached
Offline fallback page	✓ Yes	Precache on install	<code>stormo-precache-v1</code>
Push notification handler	✓ Wired	Skeleton ready	Future use
SW update detection	✓ Yes	<code>updatefound</code> → reload	Auto-updates on deploy
<code>skipWaiting</code> on install	✓ Yes	Immediate activation	New SW activates without tab close
<code>clients.claim()</code> on activate	✓ Yes	Immediate control	Existing pages adopt new SW
Old cache cleanup on activate	✓ Yes	Whitelist kept caches	Prevents stale cache growth

Section 8 → Device Test Matrix

Test in standalone mode (installed to home screen) for each platform.

8.1 — iOS

Test	Expected	Status
Install prompt shows after 4 seconds	Step-by-step guide with Share icon visible	✓ Ready
Dismissing prompt stores in <code>sessionStorage</code>	Does not reappear same session	✓ Ready
App opens in standalone mode after install	No Safari address bar, no browser chrome	✓ Ready
Status bar style	Content extends behind status bar (translucent)	✓ Ready
Dynamic Island / notch — no white band	Content extends to screen edge	✓ Ready
Home screen icon — correct brand icon	Stormo icon (180×180), no double-rounding	✓ Ready
Splash screen shows on launch	Branded dark screen with logo, no white flash	✓ Ready
Navigation works without browser back button	In-app navigation handles all routes	✓ Ready
Tap highlight removed	No blue flash on button/link tap	✓ Ready
Safe area insets (home indicator bar)	Content not hidden behind home indicator	✓ Ready

8.2 — Android / Chrome

Test	Expected	Status
Install banner appears	"Add to Home Screen" button in banner	✓ Ready
Native install prompt triggered on tap	Chrome native install sheet	✓ Ready
App opens standalone after install	No Chrome address bar	✓ Ready
Home screen icon — maskable	Fills adaptive icon shape (circle/squircle)	✓ Ready
Long-press icon — shortcuts visible	Dashboard, Content, Outreach shortcuts	✓ Ready
Theme colour in task switcher	Orange status bar (#E8621A)	✓ Ready
Offline navigation	Branded offline page shown	✓ Ready

8.3 — Desktop (Chrome / Edge)

Test	Expected	Status
Install icon in address bar	Chrome/Edge show install icon if manifest valid	✓ Ready
App opens in separate window	Standalone window, no browser chrome	✓ Ready
Install banner shown (bottom-right)	320px banner positioned bottom-right	✓ Ready
Offline page works	Branded offline page shown	✓ Ready
Pinch-to-zoom works	<code>maximumScale: 5</code> allows zoom	✓ Ready

Section 9 → Notes & Known iOS Limitations

Limitation	Detail	Mitigation
No <code>beforeinstallprompt</code> on iOS	Safari does not fire this event. Install must be done manually via Share → Add to Home Screen.	Custom step-by-step prompt in <code>PwaProvider.tsx</code>
7-day cache expiry on iOS	iOS purges PWA cache after 7 days of inactivity. This is an OS restriction.	Service worker re-caches app shell on every launch
Push notifications limited on iOS	Full push requires iOS 16.4+ and user to accept permission after install.	Push notification handler wired in SW, ready for future use
No background sync on iOS	iOS does not support Background Sync API.	Not required for current Stormo feature set
Splash screens — portrait only	Apple only uses portrait splash screens for PWA launch.	All 22 portrait screens covered; landscape files ignored
Safari-only for iOS PWA install	Chrome on iOS cannot install PWAs (Apple restriction — all iOS browsers use WebKit).	Prompt text notes "Safari only"
Status bar style <code>black-translucent</code>	Requires <code>viewport-fit=cover</code> to avoid white band at top.	Both set together in <code>layout.tsx</code>
Storage quota	iOS limits PWA storage to ~50MB shared across all PWAs.	Service worker caches only essential shell assets

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All six iOS Technical Guide requirements verified complete. Full device coverage confirmed. PWA is live and production-ready.